

ENGINEERING WORKSHOP PRACTICES

I Year I Semester: MECHANICAL ENGINEERING, AERONAUTICAL ENGINEERING & EEE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME03	ESC	L	T	P	C	CIE	SEE	Total
		0	1	2	2	40	60	100

COURSE OBJECTIVES:

1. To Study of different hand operated tools, uses and their demonstration.
2. To gain a good basic working knowledge required for the production of various engineering products.
3. To provide hands on experience about use of different engineering materials, tools, equipment's and processes those are common in the engineering field.
4. It explains the construction, function, use and application of different working tools, equipment and machines.
5. To study commonly used carpentry joints
6. Identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances.
7. To develop a right attitude, team working, precision and safety at work place

COURSE OUTCOMES:

Student will be able to:

1. Assemble different components
2. Identify and apply suitable tools for different trades of Engineering processes.
3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, House wiring, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. Apply basic electrical engineering knowledge for house wiring practice.
5. Learn the safety precautions for various operations in basic trade operations.

WEEKS	TRADES FOR EXERCISES
	Fitting
Week 1	Filing Four Sides of MS Work piece
Week 2	L Fit
	Carpentry
Week 3	Half Lap Joint
Week 4	Dove Tail Joint
	Tin Smithy

Week 5	Prepare a Rectangular Tray
Week 6	Prepare A Square Tin
	Electrical
Week 7	Series and Parallel Connection One Way Switch
Week 8	Two Way Switch Connection Stair Case Wiring
	Electronics
Week 9	Soldering - Series Connection & Parallel Connection
Week 10	Desoldering & Construction of Wheat stone bridge
WEEK	TRADES FOR DEMONSTRATION & EXPOSURE
Week 11	Introduction to Black smithy , Plumbing , Lathe Machine , Injection Moulding , Welding , Milling , Drilling and foundry.
TEXT BOOKS	
1. Work shop Manual - P. Kannaiah/ K.L. Narayana/ Scitech. 2. Workshop Practice /B. L. Juneja / Cengage 3. Rao P.N., "ManufacturingTechnology", Tata Mc Graw Hill House, Vol.I and Vol.II. 4. Workshop Manual / K. Venugopal / Anuradha Publications.	
REFERENCE BOOKS:	
1. Hajra Choudhury S.K., Hajra Choudhury A.K.and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4 th edition, 2002. 3. Workshop Manual / Venkat Reddy/ BSP	